

PMx Weekly Digest: May 29, 2026

Content Source: Full Text

Paper 1: Quantitative Evaluation of Model-Informed Drug Development Implementation in Chi

Bibliographic Information

- **Title:** Quantitative Evaluation of Model-Informed Drug Development Implementation in China's Approved Innovative Drugs: From Policy to Practice (2018–2024)
- **Authors:** Not specified in metadata
- **Journal:** Clinical Pharmacology & Therapeutics (likely, based on context), 2025
- **DOI:** N/A

Paper Type: GENERIC

Quick Take

This longitudinal analysis tracks MIDD adoption in 155 innovative drugs approved by China's NMPA from 2018–2024, documenting a surge in PopPK and E-R utilization (22% to ~88% for PopPK) following the 2020 CDE MIDD Technical Guideline. PBPK usage remained modest ($\leq 25\%$) and largely limited to DDI predictions for small molecules. By 2024, ~80% of approved drugs referenced MIDD evidence in their labels, up from 0% in 2018.

Executive Summary

This study provides the first comprehensive quantitative assessment of Model-Informed Drug Development (MIDD) implementation in innovative drugs approved by China's National Medical Products Administration (NMPA) between 2018 and 2024. Analyzing 155 chemical and biologic drugs, the authors demonstrate a dramatic increase in the use of population pharmacokinetics (PopPK) and exposure-response (E-R) analyses following the release of the 2020 CDE MIDD Technical Guideline. PopPK utilization rose from 22.2% in 2018 to 88.1% in 2024, while E-R analysis reached 85.7% in the same year. Physiologically based pharmacokinetic (PBPK) modeling, though increasingly used post-2020, remained below 25% and was predominantly applied to predict drug-drug interactions for small molecules. Crucially, the translation of MIDD evidence into approved drug labeling grew from 0% in 2018 to approximately 80% by 2024, reflecting strong regulatory acceptance. The work underscores how targeted regulatory guidance can catalyze the systematic adoption of quantitative pharmacology approaches in a rapidly evolving pharmaceutical landscape.

Scientific Context & Motivation

Model-informed drug development (MIDD) has become a cornerstone of efficient drug development globally, enabling data-driven decisions on dose selection, trial design, and labeling. While regulatory agencies such as the FDA and EMA have long encouraged MIDD, China historically lagged in its adoption. The 2020 issuance of the CDE's Technical Guidelines for Model-Informed

Drug Development marked a pivotal regulatory push to institutionalize these methodologies. Prior to this study, no systematic longitudinal assessment existed quantifying how this policy translated into actual submission and labeling practice. The knowledge gap addressed is the empirical evidence linking regulatory guidance to real-world MIDD uptake and acceptance in a major global pharmaceutical market.

✂ Methodological Snapshot

The authors conducted a retrospective longitudinal analysis of all innovative drugs (chemicals and therapeutic biologics, excluding vaccines, TCMs, and diagnostics) approved by China's NMPA from 2018 to 2024. Data were extracted from annual CDE review reports, technical review documents, labeling information at www.cde.org.cn, and select company disclosures. For each of the 155 eligible drugs, clinical pharmacology sections were systematically reviewed for evidence of MIDD approaches using predefined search terms: modeling and/or simulation, pharmacometrics, population PK (PopPK), physiologically based PK (PBPK), PK/PD modeling, disease progression modeling, model-based meta-analysis (MBMA), and exposure-response (E-R). E-R analysis was defined broadly to encompass PK/PD modeling, disease progression modeling, and MBMA. MIDD implementation was quantified as: (a) presence in regulatory submission, and (b) incorporation into approved drug labeling. Trends were analyzed by year, drug type (chemical vs. biologic), and therapeutic area (oncology vs. non-oncology). The study period captures both pre-guideline (2018–2019) and post-guideline (2020–2024) eras relative to the 2020 CDE MIDD Technical Guideline.

📋 Detailed Analysis

The paper's primary contribution is providing quantitative evidence for a phenomenon many in the pharmacometrics community have observed qualitatively: regulatory guidance drives adoption. The jump from 22% to 88% PopPK submission rate over six years is remarkable and likely reflects both increased sponsor capability and clear regulatory expectation. However, the analysis treats all MIDD applications as equivalent. A PopPK covariate analysis for label language on renal impairment is a lower-impact use than a PopPK-based dose optimization informing the pivotal dose. The 80% label acceptance rate is encouraging, but Figure 2B shows that most label inclusions for PopPK relate to ADME/covariates (foundational uses), while higher-impact applications (efficacy/safety dose adjustments) may be less common. The zanubrutinib case study illustrates best practice but may not represent the median submission. The authors acknowledge that companies may perform internal high-impact MIDD analyses (e.g., dose selection) without submitting them—a missed opportunity. This 'submission gap' is an important finding: even as adoption grows, conservative submission strategies may limit the full potential of MIDD to influence regulatory decisions. The modest PBPK utilization ($\leq 25\%$, mostly DDI) aligns with global trends where PBPK is accepted for DDI but faces higher hurdles for other applications (e.g., formulation effects, pediatric extrapolation). The absence of QSP or AI/ML submissions, despite in-house use, highlights a 'regulatory innovation lag' that the ICH M15 guideline may help address.

Domain Context

This paper sits at the intersection of regulatory science, pharmacometrics policy, and pharmaceutical industry practice. It complements similar analyses from the FDA (e.g., MIDD-related labeling decisions, impact on approval) and PMDA (Japan), but uniquely focuses on China's rapidly evolving regulatory environment. The findings are particularly relevant given China's growing role in global drug development and the international harmonization efforts through ICH. The adoption trajectory observed (PopPK > E-R > PBPK > advanced methods) mirrors the maturity and regulatory acceptance hierarchy seen globally. The 2020 CDE guideline was preceded by a 2018 FDA MIDD pilot program and 2020 PMDA MIDD guidance, placing China's timeline in a global context of increasing regulatory emphasis on quantitative approaches. The study does not address the role of MIDD in regulatory flexibility (e.g., waiving dedicated studies, supporting extrapolation), which is an important dimension of impact.

☒ Key Findings

- 1) PopPK and E-R analysis utilization in NMPA submissions grew steadily from 22.2% (2018) to 88.1% and 85.7% (2024), respectively.
 - 2) PBPK modeling utilization emerged post-2020 but remained $\leq 25\%$, nearly exclusively for chemical drugs (15/16 applications) and focused on DDI predictions.
 - 3) Incorporation of MIDD evidence into approved drug labeling rose from 0% in 2018 to $\sim 50\%$ by 2021 and $\sim 80\%$ by 2024, with comparable rates between chemicals and biologics.
 - 4) Among labeled MIDD data, PopPK was most common (53.5% of 155 drugs), while E-R relationships (3.9%) and PBPK (8.4%) were less frequently documented.
 - 5) Subgroup analyses showed comparable PopPK/E-R adoption between chemical and biologic drugs (2021–2024), and earlier preferential oncology adoption before 2022.
 - 6) The zanubrutinib case exemplifies successful multi-MIDD approach integration supporting dose bridging, DDI management, and labeling.
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Strengths & Limitations

Strengths

- First longitudinal quantitative analysis of MIDD implementation in China covering pre- and post-guideline eras (2018–2024)
- Systematic review of regulatory dossiers, technical review reports, and labeling from official CDE database
- Clear categorization of three major MIDD approaches (PopPK, E-R, PBPK) with defined inclusion criteria
- Provides concrete metrics (submission rates, labeling acceptance rates) that document policy impact
- Includes an instructive case study (zanubrutinib) illustrating successful integrated MIDD application

Limitations (Acknowledged by Authors)

- Increasing MIDD adoption is shaped by multiple factors beyond specific NMPA guidance (global innovative drug applications, ICH E17, methodological advancements, successful case studies)
- Exclusion of vaccines, traditional Chinese medicines, and diagnostic reagents
- Reliance on publicly accessible documentation which may not capture all internal MIDD analyses not submitted to regulators

Limitations (Expert Review)

- Causal attribution of the MIDD increase to the 2020 guideline is correlational, not causal—confounding with global trends and industry maturation is likely
- No analysis of the quality or impact level of MIDD analyses (e.g., high-impact dose optimization vs. routine covariate analysis) beyond labeling presence
- Sample size for subgroup comparisons (e.g., oncology vs. non-oncology PBPK) is limited, precluding definitive conclusions
- The study does not assess whether MIDD adoption reduced development timelines or costs, which would strengthen the policy impact narrative
- The 2024 data likely includes drugs approved very recently; long-term label acceptance follow-up may be incomplete

Generalizability

The findings are directly generalizable to China’s pharmaceutical regulatory ecosystem, and the observed adoption patterns are consistent with reports from PMDA (Japan). However, direct extrapolation to FDA or EMA contexts requires caution given differences in regulatory cultures and guidance timelines. The zanubrutinib case suggests that Chinese-origin global drugs may serve as benchmarks, but broader generalizability to all therapeutic areas and drug modalities is supported by the comprehensive sampling across 7 years.

Figures & Tables

- **Figure 1:** (A) Bar chart showing number and categories of innovative drugs approved by NMPA (2018–2024), with total of 195 drugs. (B) Line chart showing percentage of approved drugs utilizing PopPK modeling, E-R analysis, and PBPK modeling over the same period.
 - *Significance:* Provides the core longitudinal data demonstrating the dramatic increase in PopPK and E-R adoption from ~22% to ~88%, and the modest, variable uptake of PBPK.
 - **Figure 2:** (A) Bar chart showing percentage of approved drugs with MIDD-derived data documented in labeling (2018–2024), rising from 0% to ~80%. (B) Breakdown of label statements supported by PopPK or PBPK modeling, categorized by type (ADME, covariates, dose adjustments, DDI, food effects).
 - *Significance:* Demonstrates regulatory acceptance of MIDD evidence through labeling impact, and shows the types of decisions supported by each modeling approach.
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Code & Reproducibility Assessment

No code or data repository is mentioned. The analysis is based on publicly available CDE review reports and labeling, but the raw extraction and classification methodology is described qualitatively rather than as a reproducible script.

Supplementary Materials

Table S1 (referenced) provides subgroup analysis of PopPK and E-R utilization rates comparing chemical drugs vs. biologics, and oncology vs. non-oncology, but the content was not available in the provided text.

Future Directions

1) Evaluate the impact level of submitted MIDD analyses (high-stakes decisions like pivotal dose selection vs. routine covariate analysis) to better characterize maturity. 2) Conduct similar analyses for vaccines and traditional Chinese medicines as MIDD adoption expands to these modalities. 3) Assess development timelines, cost savings, and regulatory success rates associated with MIDD-informed programs. 4) Track the adoption of emerging methods (QSP, AI/ML, model-based meta-analysis) as regulatory comfort grows and the ICH M15 guideline is implemented. 5) Perform cross-agency comparisons (NMPA vs. FDA vs. EMA vs. PMDA) on same-drug submissions to disentangle global versus local regulatory effects.

Expert Commentary

This paper provides welcome empirical evidence that regulatory guidance can effectively drive adoption of quantitative methods. The near-90% PopPK submission rate by 2024 is impressive and mirrors trends seen in mature regulatory agencies. However, the 80% labeling acceptance rate suggests that regulators are not merely receiving these analyses but are acting on them, which is the more important metric. The low PBPK utilization outside DDI is notable and likely reflects both the complexity of building and validating these models and a conservative regulatory approach. As the ICH M15 guideline harmonizes MIDD expectations globally, we should expect Chinese companies to increasingly submit higher-impact modeling work—including for dose optimization and pediatric extrapolation—that they may currently reserve for internal decisions. The paper would have been strengthened by analyzing the ‘impact level’ of each submission rather than just presence/absence.

Bottom Line

China’s 2020 MIDD Technical Guideline has successfully catalyzed widespread adoption of PopPK and E-R analyses in innovative drug submissions, with regulatory acceptance reflected in labeling. However, PBPK and advanced methods (QSP, AI) remain underutilized in submissions, representing an opportunity for future growth. Practitioners developing drugs for the Chinese market should anticipate regulatory expectations for MIDD evidence, particularly PopPK and E-

R, and should consider submitting internal modeling work even for high-impact decisions to build the regulatory track record.

☒ **Figures**

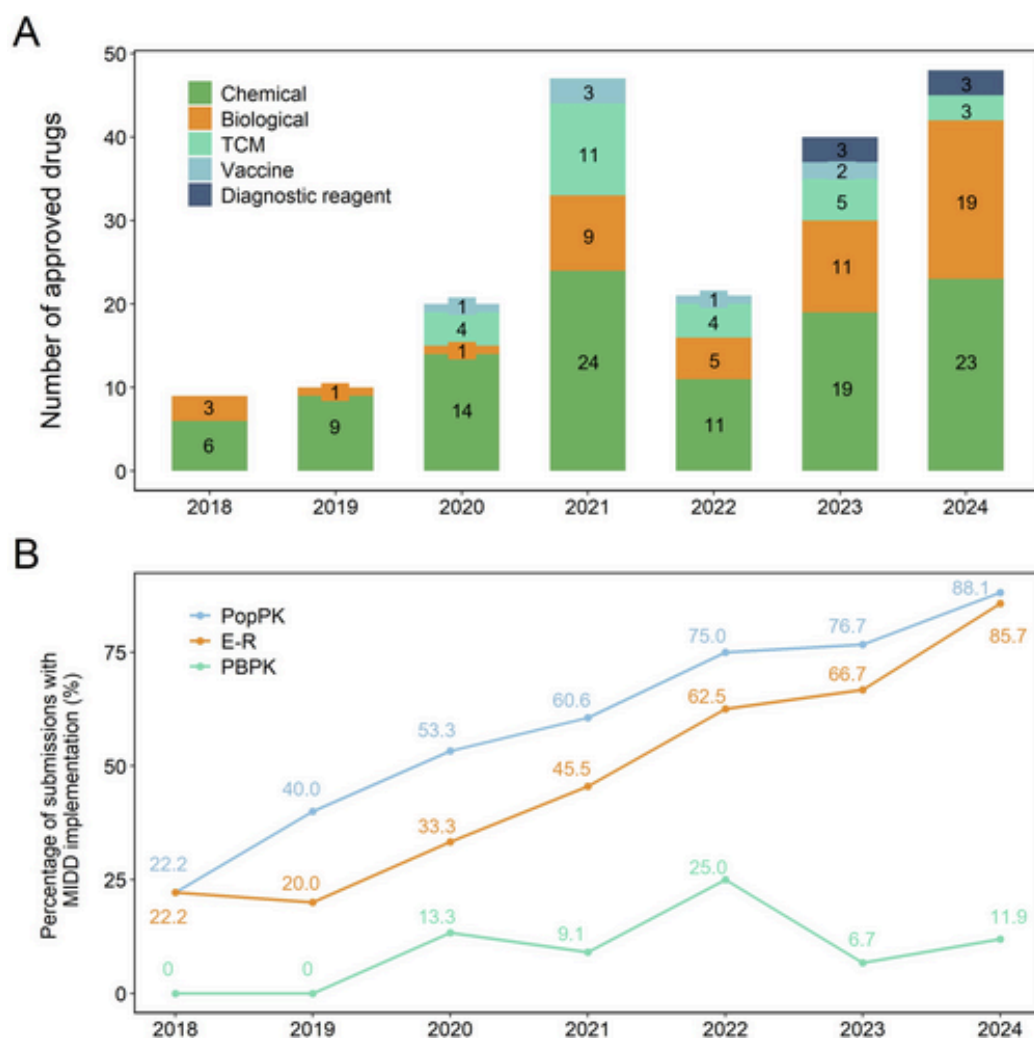


Figure 1: Innovative drug approvals and MIDD utilization patterns in China. (A) Number and categories of innovative drugs approved by the NMPA of China from 2018 to 2024.

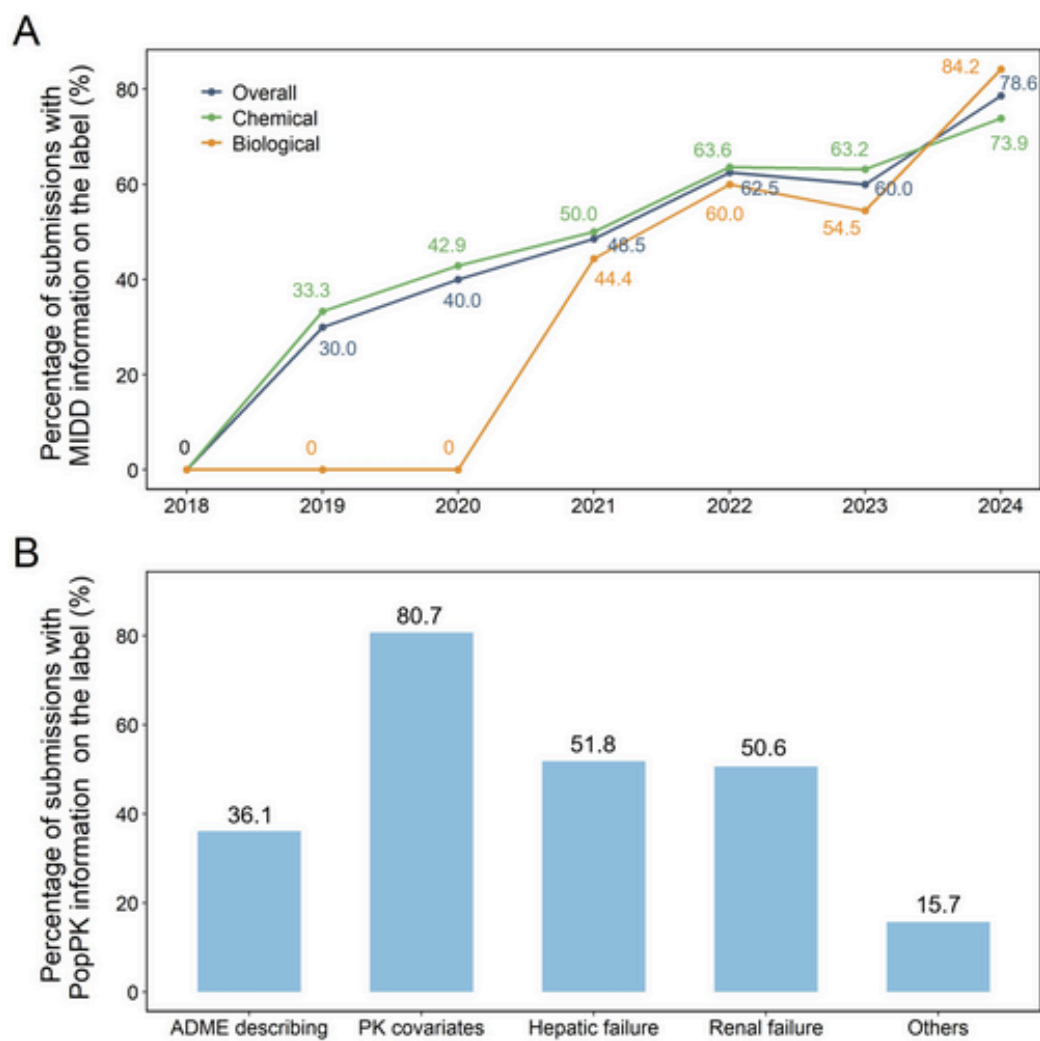


Figure 2: Integration of MIDD information in drug labeling in China. (A) Percentages of approved drugs with MIDD-derived data explicitly documented in labeling (2018–2024)